JULES

Six linked slopes: envelopes, LFOs, oscillators, chords

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Jules is six slope generators that share one set of controls. RATE sets how fast the first one goes; SPREAD sets how the other five relate to it: all the same, or 2, 3, 4, 5 and 6 times faster, or slower.

Depending on the range and the mode, the six slopes become cascading envelopes, polyrhythmic LFOs, six slew limiters, a chord of oscillators, plucked notes or FM voices. CHANCE adds controlled randomness to the way triggers travel across the six.

AT A GLANCE

- 6 slopes sharing RATE, SPREAD, RISE and CURVE
- CV range (envelopes and LFOs, 0 to 8 V) or AUDIO range (oscillators, ±5 V)
- 12 modes on one selector, each one explained on the screen
- Triggers cascade from right to left, with a CHANCE control
- Six outputs, a mix and a 6-channel poly output
- 18 HP

CONTROLS

Screen	Seven bars (the six slopes and the mix), the mode name, and what TWEAK does in that mode.
RATE	Speed of slope 1. CV range: from about 2 minutes to 8 ms per cycle, 1 Hz at the centre. AUDIO range: about 12 Hz to 6 kHz, middle C at the centre.
SPREAD	Speed of slopes 2 to 6 compared with slope 1. Centre: all the same. Right: slope N goes up to N times faster (fully right ×2, ×3 … ×6, a harmonic series). Left: up to N times slower (an undertone series). Finer near the centre.
FM	Amount of the FM input. Right of centre: modulates the speed of all six slopes equally. Left of centre: modulates SPREAD, so slope 1 is untouched and the fundamental stays solid. Centre: off.
RISE	Share of each cycle spent rising, without changing its length. Left: instant rise, long fall (saw). Centre: triangle. Right: long rise, instant fall (ramp).
CURVE	Bends the slopes without changing their timing. Centre: straight lines. Right: exponential, then sine. Left: logarithmic, then square; fully left, RISE becomes the pulse width.
CV / AUDIO	Range switch. CV: slow, outputs 0 to 8 V. AUDIO: fast, outputs ±5 V.
MODE	Six positions, each with a CV and an AUDIO version: see the mode table.
TWEAK	A second control for modes 2, 4 and 6. The small light shows when it is active; the screen says what it does.
CHANCE	Probability that a trigger passed along from the right reaches each slope. 100 % (default): every slope fires. 50 %: each one flips a coin on every trigger. Triggers patched directly always pass.

MODES

#	CV range	AUDIO range
1	**ONE-SHOT**. Each trigger draws one rise and fall. Triggers that arrive while a slope is moving are ignored.	**IMPULSE**. The same at audio rate: patch an audio-rate clock into TRIG for impulse-train oscillators.
2	**RETRIGGER**. Like ONE-SHOT, but TWEAK sets from when a moving slope accepts a new trigger: −5 V at any time, 0 V after its peak, +5 V only once finished. It then restarts from zero.	**SUBHARMONIC**. Slope 1 runs freely and fires the other five at the end of each of its cycles. TWEAK sets when they accept it. With SPREAD left: subharmonics, split tones and sync chaos.
3	**GATE**. Rises while the gate is high, holds at the top, falls when the gate drops.	**PULSE**. The same with an audio-rate clock: oscillators whose shape follows the clock's pulse width.
4	**SUSTAIN**. Rises, falls to a sustain level set by TWEAK (−5 V: 0 %, +5 V: 100 %), holds while the gate is high, then falls. With a steady gate, the six outputs follow TWEAK at six speeds: six slew limiters.	**PLUCK**. Six oscillators, each through its own lowpass gate. Triggers pluck notes, gates hold them. TWEAK sets the decay: about 300 ms at the centre, shorter to the right, longer to the left.
5	**LOOP**. Six free-running LFOs. Triggers reset them to the start of their cycle.	**OSC**. Six oscillators; triggers hard-sync them. SPREAD fully right gives a major chord, fully left a minor chord, across a few octaves.
6	**BURST**. Slopes rest until triggered, then run a burst of cycles: TWEAK −4 V: 1, 0 V: 6, +5 V: 36. Below −4 V triggers are ignored.	**FM**. Each oscillator gets its own sine modulator. TWEAK sets its ratio (×0.5 at −5 V, ×1 at 0 V, ×2 at +5 V), the FM knob the depth (right: equal, left: more on the higher slopes), the FM input scales it.

INPUTS

V/OCT	Added to RATE, 1 volt per octave: the speed doubles with each volt.
SPREAD · RISE · CURVE	Added to their knobs; ±5 V covers the whole range.
FM	Frequency modulation source. In the AUDIO range a DC blocker keeps the pitch centred. In FM mode it becomes a depth control.
TWEAK	Added to the TWEAK knob, ±5 V.
CHANCE	Added to the CHANCE knob; 10 V = 100 %.
TRIG 1 to 6	Trigger or gate for each slope, threshold about 1 V. Unpatched inputs take the signal from the right (see the trigger cascade).

OUTPUTS

1 to 6	Each slope: 0 to 8 V in the CV range, ±5 V in the AUDIO range.
MIX	CV range: the highest of the six slopes, each divided by its number. AUDIO range: the six summed and gently limited, about ±7.5 V.
POLY	The six slopes in one 6-channel cable.

HOW IT WORKS

The trigger cascade

Each TRIG input that has no cable receives the signal of the nearest patched input to its right. One cable into TRIG 6 therefore drives all six slopes; a second cable into TRIG 3 splits them into two groups (6, 5, 4 and 3, 2, 1).

CHANCE acts on that cascade only: every time a trigger travels to a slope through the cascade, the slope decides at random whether to take it. This is the quickest way to make six voices play a different combination on every beat while staying in time.

The MIX output

In the CV range, MIX outputs the highest of the six slopes, each divided by its number (slope 3 at a third of its level, and so on): a complex modulation that would otherwise need a whole patch. In the AUDIO range it is simply the six summed and gently limited.

PATCH IDEAS

1. Self-playing chords

AUDIO range, MODE on PLUCK, SPREAD fully right. A clock into TRIG 6, CHANCE around 60 %. MIX into a delay and a reverb: each beat plucks a different handful of chord notes.

2. Polyrhythmic LFOs

CV range, LOOP, SPREAD a little right of centre. Six LFOs drift in and out of phase; send them to filters, pans and levels.

3. Cascading envelopes

CV range, ONE-SHOT, SPREAD to the left. One trigger into TRIG 6 gives six envelopes of growing length.

4. Six slews from one random line

CV range, SUSTAIN, a steady +10 V into TRIG 6 and a stepped random voltage into the TWEAK input: six smoothed versions of the same line, from sharp to lazy. Quantize a few of them for harmonies that arrive at different times.

5. FM bass

AUDIO range, FM mode. A sequence into V/OCT, an envelope into the FM input, use output 1.

SPECIFICATIONS

Width	18 HP
CV range	Slope 1 from about 0.008 Hz to 128 Hz (up to ×6 with SPREAD)
AUDIO range	Slope 1 from about 12 Hz to 6 kHz, 1 V/oct
Levels	Outputs 0 to 8 V (CV) or ±5 V (AUDIO). Trigger threshold about 1 V. POLY: 6 channels.